

Project Design Phase

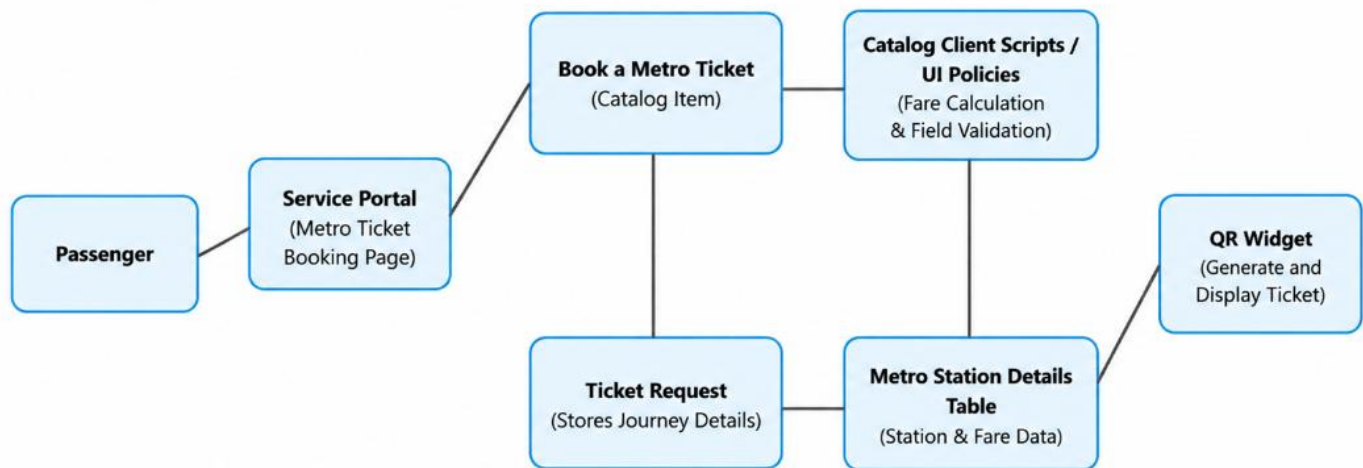
Solution Architecture

Date	04 September 2026
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Project Name	Metro Ticket Generating System
Maximum Marks	

Solution Architecture

The solution architecture connects the **passenger-facing ServiceNow portal** with the metro ticket booking configuration, fare calculation logic, payment handling, QR generation, and backend station and ticket data. The architecture is designed to provide a guided booking process while maintaining accurate ticket information.

Solution Architecture – Metro Ticket Generating System



Passenger → Service Portal → Book a Metro Ticket → Ticket Request → Rules → Data → QR Widget

Figure 1: Employee Raise Issue solution architecture.

Layer	Components	Responsibility
Experience	Service Portal, Metro Ticket Booking page, QR Widget	Provides the passenger-facing booking experience.
Interaction	Catalog Client Scripts	Handles dynamic form behaviour and automatic fare calculation.
Catalog	Book a Metro Ticket Catalog Item, Variables	Collects journey, passenger, fare, and payment information.
Rules	UI Policies, dependent choices, validation logic	Controls field visibility, mandatory fields, and valid selections.
Data	Metro Station Details table, Ticket/Request data	Stores station information and submitted ticket details.
Integration / Output	QR Widget, QR generation logic	Generates and displays the digital QR ticket after submission.